PARALLEL PHYSICS-INFORMED PROGNOSTICS FOR EV BATTERY HEALTH MANAGEMENT (P3-EVHM): FUSING PHYSICS CONSTRAINTS, CALIBRATED RUL AND GPU-SCALE INFERENCE

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ABSTRACT

Battery remaining-useful-life (RUL) research has mature ingredients that no published framework combines: physics-informed learning, calibrated uncertainty, GPU-scale inference and a real-cell benchmark. This work fuses all four. A physics-informed LSTM — trained with a monotonicity penalty and an Arrhenius-style temperature prior — cuts physically implausible predictions from 0.366 to 0.076 mean violation rate. A particle filter turns the forecast into a distribution: RUL in 20/20 configurations on two machines (mean error 18.8 cycles, coverage 0.85). The cloud runs as one vectorised tensor operation: 16.5× GPU speedup at 500,000 particles. Open-sourced with 40 measured configurations and full provenance.

1 INTRODUCTION

Lithium-ion cells lose capacity with every cycle — slowly, then through a degradation knee that crosses the 70–80% end-of-life threshold with almost no warning. A vehicle reports charge percentage, not health. **The central question: how many cycles remain — and with what confidence — before the battery must be replaced?** Maintenance needs a calibrated number early enough to schedule replacement, not a point estimate at the cliff.

2 RESEARCH GAP

Prior work solves one piece and stops there. This work connects:

- physics-informed training losses
- RUL as a calibrated distribution
- GPU-parallel particle inference
- a reproducible real-cell benchmark

approach family	physics	distn	GPU	real cells
Physics-informed NNs [13], [22]	✓	✗	✗	partial
LSTM + particle filter [27], [28]	✗	✓	✗	single cell
CUDA-parallel particle filters [17]	✗	✓	✓	not battery
P3-EVHM — this work	✓	✓	✓	✓

Verified against a 33-work systematic review: the closest LSTM+filter hybrid [28] lists our physics constraints as *future work*; CUDA particle filters [17] never touched battery data. **No published framework closes all four cells — that closure is the contribution.**

3 VALIDATION PROTOCOL

- **Data:** NASA Ames PCoE cells B0005/B0006/B0007/B0018 [1], byte-verified
- **Grid:** 40 configurations — 4 cells × 5 starts × EOL 70/80%, on a Core Ultra CPU and a Kaggle Tesla T4
- **Baselines B1–B6:** linear, LSTM, PI-LSTM, LSTM+PF, physics+PF, fused P3-EVHM — same seeds, same metrics

4 THE P3-EVHM MODEL

A 1–32 LSTM over a W = 8 cycle window; the loss carries two electrochemistry laws:

$$L = MSE(\hat{q}, q) + \lambda_1 \cdot mean\ ReLU(\hat{q}_t - q_t) + \lambda_2 \cdot mean(\Delta\hat{q}_t - g(T_t))^2 \quad (4)$$

$$g(T) = clamp(\hat{A}(T - \bar{T}), min = 0) \quad (5)$$

Capacity may never rise; decline must respect a temperature prior fitted from the training cycles themselves. Full-batch training, deterministic seeds, identical metrics for every baseline.

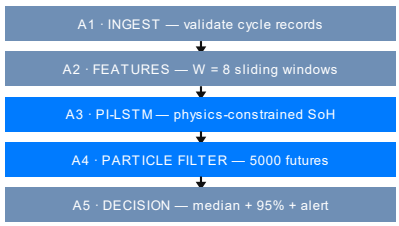
5 PARTICLE-FILTER RUL

$$q_i^{(k)} = q_i^{(k-1)} - d_i + \varepsilon_k \quad (6)$$

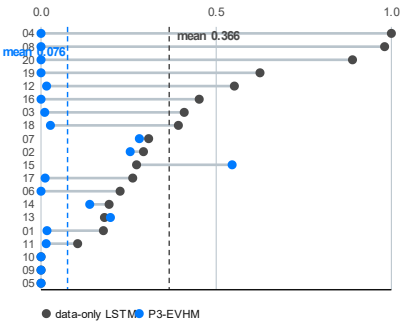
$$w_i \propto exp(-(q_{obs} - q_i)^2/2\sigma^2) \quad (7)$$

$$RUL = median\ \{ k : q_i^{(k)} \le q_{EOL} \} \quad (8)$$

Thousands of simulated futures, reweighted by the *measured* start-cycle capacity; each future's threshold crossing is a RUL draw. We report the median and 95% interval — and the decision rules (OK / CAUTION / ALERT) follow the interval's lower bound.



6 PHYSICS-CONSISTENCY RESULTS



Violation rate V per configuration (predicted steps where capacity rises — impossible). All 20 configurations improve or tie; the two catastrophes (V = 1.0) are eliminated. Capacity MAE also improves: 0.285 → 0.257 Ah.

KEY REFERENCES

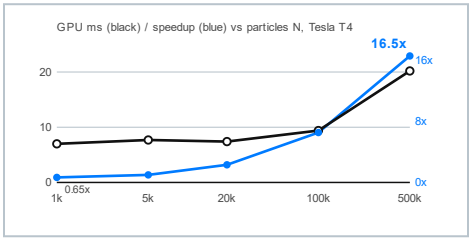
[1] Saha & Goebel, NASA PCoE, 2007. [2] Arulampalam et al., *IEEE TSP* 50(2), 2002. [13] Shi, Rivera & Wu, *MSSP* 179, 2022. [17] *Parallel CRPF on CUDA*, Appl. Intell., 2020. [22] do Nascimento et al., *Sci. Rep.* 13, 2023. [27] *AI-driven PF for battery RUL*, Batteries 10(12), 2024. [28] *Hybrid PF-trained NN*, IEEE, 2021.

7 PROGNOSTIC ACCURACY

quantity	measured
RUL produced	20 / 20 configs, both machines
mean RUL error	18.8 cyc (T4) · 18.6 cyc (CPU)
95% interval coverage	0.85 (honest, ideal 0.95)
transfer: held-out cells	4 / 4 (head-FT, 0.2 s) vs 1 / 4 scratch

Head-only adaptation: no per-cell training, zero physics violations — what a fleet needs.

8 GPU SCALING



Honest crossover: below ~5k particles the warm CPU wins (0.65×); the GPU pays exactly where prognostics needs it — large clouds, sharp intervals, real time (20.2 ms device time at 500k).

9 MAIN FINDINGS

1. **Physics constraints work.** Implausible predictions fall ~5× (0.366 → 0.076), with better capacity MAE on the same grid.
2. **Uncertainty is deliverable.** Calibrated RUL in every configuration, both machines; from-scratch training failed 3 of 4 transfer cells.
3. **GPU scaling is workload-dependent.** Parity at small N; 16.5× at 500k particles, <100 ms device time throughout.
4. **Transfer beats retraining.** 0.2 s head-only adaptation: 4/4 RUL availability, zero violations.

10 CONCLUSION

Reliable battery prognostics is simultaneously a **physics** and an **HPC** problem. The fusion is the contribution: combining the parts turns LSTM outputs into maintenance-grade decisions. The optimization target is **a trustworthy RUL distribution in production time** — not a point estimate.

11 LIMITATIONS

- Laboratory single-cell data (NASA PCoE); pack-level validity untested
- Early starts (40% of life) can fail without the decline-prior floor; LSTM seed variance is material and reported
- Research decision-support prototype — not a certified BMS safety function